CIA 2

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1. Objective and Scope

- **Primary Objective:** Design and implement a fully automated Continuous Integration and Continuous Deployment (CI/CD) pipeline for a Java Spring Boot application.
- **Scope:** Fix a known boundary condition defect (Set E), containerize the application, orchestrate the deployment using Kubernetes, and automate the entire workflow utilizing Jenkins.

2. Development Environment and Toolchain

- **Operating System:** Windows 11
- **Runtime Environment:** Java 17
- **Build Tool:** Apache Maven 3.9
- **Containerization:** Docker Desktop
- **Orchestration:** Kubernetes (Minikube local cluster)
- **Automation Server:** Jenkins (running as a local Windows service)
- **Version Control:** Git

3. Version Control and Branching Strategy

- **Feature Isolation:** Created a dedicated branch (feature/positive-quantity-validation) to isolate defect resolution and pipeline configurations from the main codebase.
- **Commit Strategy:** Utilized atomic commits to track distinct phases of development, including defect fixing, manifest creation, and pipeline automation.
- **Working Tree Verification:** Maintained a clean working tree prior to deployment to ensure code integrity.

4. Defect Analysis and Resolution (Set E)

- **4.1 Identification of the Boundary Condition Flaw:** The initial BookingService.java logic exhibited an off-by-one error, incorrectly approving requests where the quantity requested was exactly one unit greater than the available inventory.
- **4.2 Implementation of Positive Quantity Validation:** Introduced strict boundary enforcement (requestQuantity > availableInventory) and input sanitization guard clauses (requestQuantity <= 0) to actively reject zero or negative booking requests, preventing inventory state corruption.
- **4.3 Automated Boundary Testing via JUnit:** Developed and executed targeted unit tests to validate the newly established thresholds. All 9 tests passed successfully, confirming robust input validation.

5. Application Build and Packaging (Maven)

- **Lifecycle Execution:** Executed mvn clean package to compile source code, execute the test suite, and package the application.
- **Artifact Generation:** Produced a deployable, standalone executable JAR file configured with embedded server dependencies.

6. Containerization Strategy (Docker)

- **6.1 Dockerfile Configuration:** Utilized eclipse-temurin:17-jre-alpine as a lightweight base image to minimize surface area and footprint. Exposed port 8080 and defined the entry point for the compiled JAR.
- **6.2 Image Build and Verification:** Successfully built and tagged the image as campus-equipment-booking:set-e, subsequently verifying its presence and size within the local Docker registry.

7. Container Orchestration (Kubernetes / Minikube)

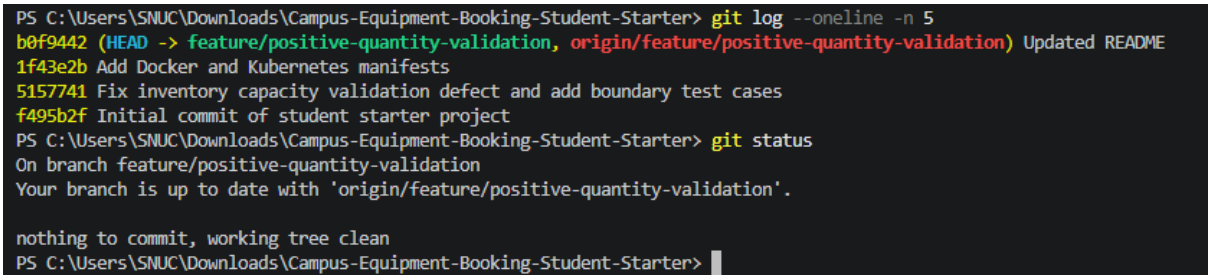
- **7.1 Deployment Configuration:** Defined declarative YAML manifests to manage the application lifecycle and network services. Loaded the local Docker image directly into the Minikube environment.
- **7.2 Pod Scaling and High Availability:** Scaled the deployment to two replicas (replicas=2) to demonstrate load distribution and fault tolerance.
- **7.3 Service Exposure and Port Forwarding:** Mapped the internal cluster service to localhost via kubectl port-forward. Due to a port conflict with the Jenkins host on 8080, traffic was successfully routed to external port 8081.
- **7.4 Health Check Verification:** Validated application readiness by successfully polling the Spring Boot Actuator endpoint (/actuator/health), returning a 200 OK status.

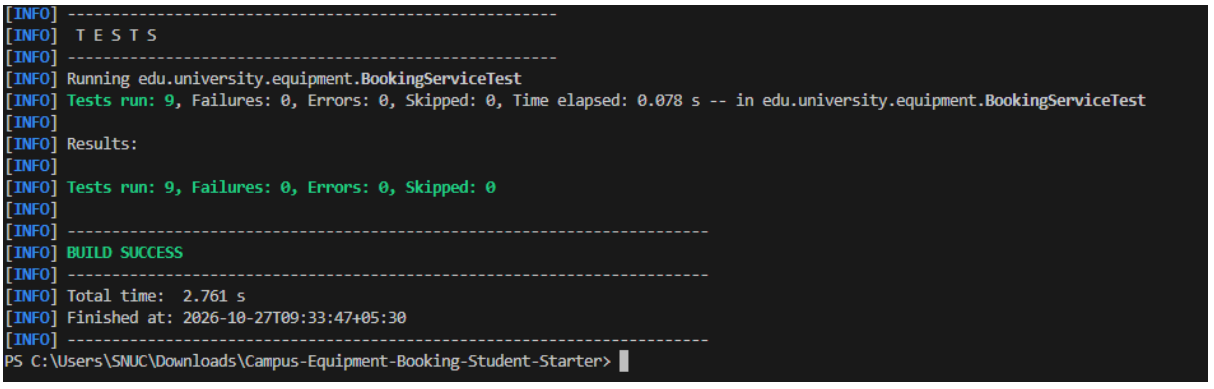
8. Continuous Integration and Deployment [CI/CD] (Jenkins)

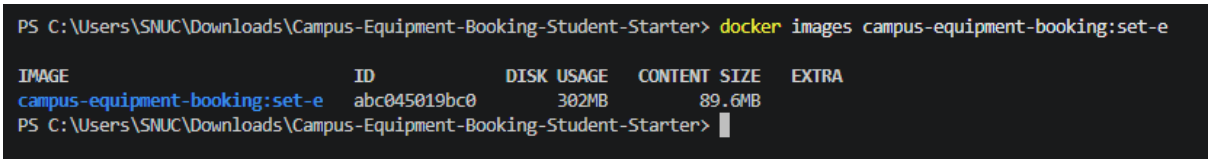
- **8.1 Pipeline Architecture:** Drafted a Groovy-based declarative Jenkinsfile consisting of three distinct execution stages: Build & Test, Docker Build, and Deploy to Kubernetes.
- **8.2 Environment Configuration:** Resolved critical integration hurdles by removing UTF-16 BOM encoding from the pipeline script, bypassing default Jenkins local Git checkout security restrictions, and reconfiguring the Jenkins Windows service to run under a user profile with Docker and Minikube access.
- **8.3 Automated Execution Workflow:** Achieved end-to-end automation where code commits trigger a complete pipeline execution, resulting in successful container deployment without manual terminal intervention.

9. Final Output and Evidence

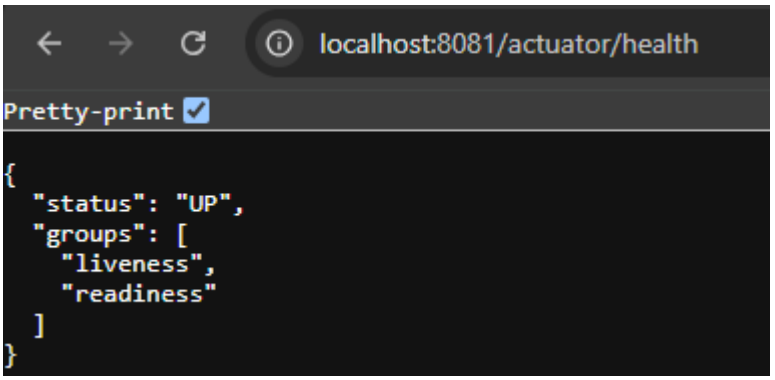
- **Documentation:** Updated the project README.md to reflect build commands and defect resolutions.
- **Visual Evidence:** Captured and version-controlled screenshots validating the Git history, Maven build success, Docker image creation, Kubernetes pod scaling, and the successful Jenkins pipeline console output.



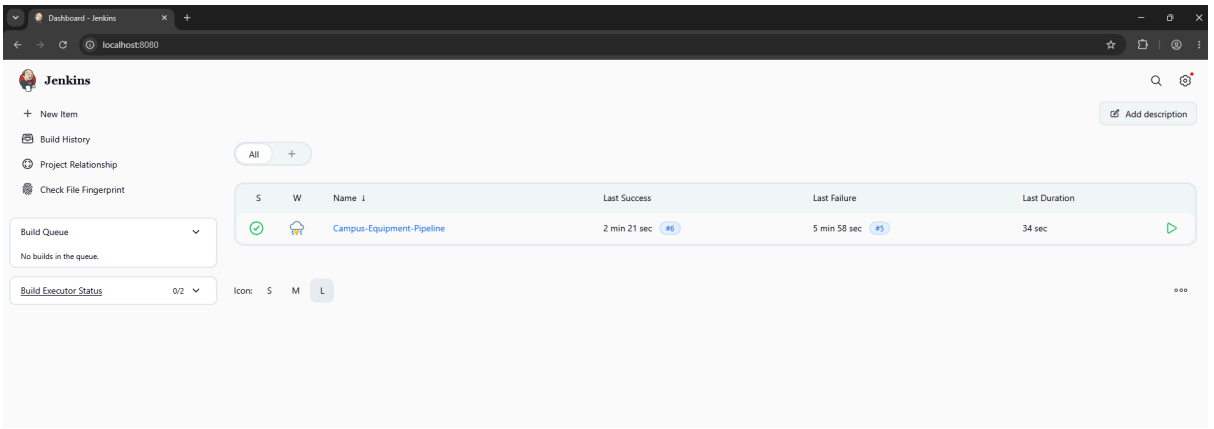




```
PS C:\Users\SNUC\Downloads\Campus-Equipment-Booking-Student-Starter> kubectl get deployments
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
campus-equipment-deployment   2/2     2            2           17m
PS C:\Users\SNUC\Downloads\Campus-Equipment-Booking-Student-Starter> kubectl get pods
NAME                                          READY   STATUS    RESTARTS   AGE
campus-equipment-deployment-6c59d694c6-8z68j   1/1     Running   0          18m
campus-equipment-deployment-6c59d694c6-w24p5   1/1     Running   0          16m
PS C:\Users\SNUC\Downloads\Campus-Equipment-Booking-Student-Starter>
```




```
Forwarding from 127.0.0.1:8081 -> 8080
Forwarding from [::1]:8081 -> 8080
Handling connection for 8081
Handling connection for 8081
Handling connection for 8081
Handling connection for 8081
Handling connection for 8081
Handling connection for 8081
```



```
C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] --------------< edu.university:campus-equipment-booking >---------------
[INFO] Building campus-equipment-booking 1.0.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.3.2:clean (default-clean) @ campus-equipment-booking ---
[INFO] Deleting C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ campus-equipment-booking ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ campus-equipment-booking ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 6 source files with javac [debug parameters release 17] to target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ campus-equipment-booking ---
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ campus-equipment-booking ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ campus-equipment-booking ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.university.equipment.BookingServiceTest
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.063 s -- in edu.university.equipment.BookingServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ campus-equipment-booking ---
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline\target\campus-equipment-booking-1.0.0.jar
[INFO]
[INFO] --- spring-boot:3.3.5:repackage (repackage) @ campus-equipment-booking ---
[INFO] Replacing main artifact C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline\target\campus-equipment-booking-1.0.0.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline\target\campus-equipment-booking-1.0.0.jar.original
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.978 s
[INFO] Finished at: 2026-10-27T10:03:58+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Docker Build)
[Pipeline] bat
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline>docker build -t campus-equipment-booking:set-e .
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 199B 0.0s done
#1 DONE 0.1s

#2 [internal] load metadata for docker.io/library/eclipse-temurin:17-jre-alpine
#2 DONE 0.1s

#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.1s

#4 [1/3] FROM docker.io/library/eclipse-temurin:17-jre-alpine@sha256:c4a33e6ddb738fe4be0a28a613ea0940cec42d25d7263c3e2238c43354cb19b0
#4 resolve docker.io/library/eclipse-temurin:17-jre-alpine@sha256:c4a33e6ddb738fe4be0a28a613ea0940cec42d25d7263c3e2238c43354cb19b0 0.1s done
#4 DONE 0.1s

#5 [internal] load build context
#5 transferring context: 24.18MB 1.8s done
#5 DONE 1.9s

#6 [2/3] WORKDIR /app
#6 CACHED

#7 [3/3] COPY target/campus-equipment-booking-1.0.0.jar app.jar
#7 DONE 0.1s

#8 exporting to image
#8 exporting layers
#8 exporting layers 0.6s done
#8 exporting manifest sha256:5e2af3df307d7f4dfed24d9325812a290dbd95e479f90cd6d4eb8966c64b5bba 0.0s done
#8 exporting config sha256:62bd338759eac808e2b558544e99c2da74676156c8682bda8bc32fbcf884dafc 0.0s done
#8 exporting attestation manifest sha256:1c423a4114e075027b5e1bb9280e736504d5157a63b09d6c0fee32757d533fe6 0.0s done
#8 exporting manifest list sha256:2d900f91f8d228c29e17ee0934fb91580b49ca4f2535f42520f696428607d3e7
#8 exporting manifest list sha256:2d900f91f8d228c29e17ee0934fb91580b49ca4f2535f42520f696428607d3e7 0.0s done
#8 naming to docker.io/library/campus-equipment-booking:set-e done
#8 unpacking to docker.io/library/campus-equipment-booking:set-e
#8 unpacking to docker.io/library/campus-equipment-booking:set-e 0.2s done
```


```
C:\ProgramData\Jenkins\.jenkins\workspace\Campus-Equipment-Pipeline>docker build -t campus-equipment-booking:set-e .
#0 building with "desktop-linux" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 199B 0.0s done
#1 DONE 0.1s

#2 [internal] load metadata for docker.io/library/eclipse-temurin:17-jre-alpine
#2 DONE 0.1s

#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.1s

#4 [1/3] FROM docker.io/library/eclipse-temurin:17-jre-alpine@sha256:c4a33e6ddb738fe4be0a28a613ea0940cec42d25d7263c3e2238c43354cb19b0
#4 resolve docker.io/library/eclipse-temurin:17-jre-alpine@sha256:c4a33e6ddb738fe4be0a28a613ea0940cec42d25d7263c3e2238c43354cb19b0 0.1s done
#4 DONE 0.1s

#5 [internal] load build context
#5 transferring context: 24.18MB 1.8s done
#5 DONE 1.9s

#6 [2/3] WORKDIR /app
#6 CACHED

#7 [3/3] COPY target/campus-equipment-booking-1.0.0.jar app.jar
#7 DONE 0.1s

#8 exporting to image
#8 exporting layers
#8 exporting layers 0.6s done
#8 exporting manifest sha256:5e2af3df307d7f4dfed24d9325812a290dbd95e479f90cd6d4eb8966c64b5bba 0.0s done
#8 exporting config sha256:62bd338759eac808e2b558544e99c2da74676156c8682bda8bc32fbcf884dafc 0.0s done
#8 exporting attestation manifest sha256:1c423a4114e075027b5e1bb9280e736504d5157a63b09d6c0fee32757d533fe6 0.0s done
#8 exporting manifest list sha256:2d900f91f8d228c29e17ee0934fb91580b49ca4f2535f42520f696428607d3e7
#8 exporting manifest list sha256:2d900f91f8d228c29e17ee0934fb91580b49ca4f2535f42520f696428607d3e7 0.0s done
#8 naming to docker.io/library/campus-equipment-booking:set-e done
#8 unpacking to docker.io/library/campus-equipment-booking:set-e
#8 unpacking to docker.io/library/campus-equipment-booking:set-e 0.2s done
```